

WRIGHT-PATTERSON AFB, OH, USA

WMO: 745700

Lat: 39.833N

Lon: 84.050W

Elev: 251

StdP: 98.35

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13840

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-13.0	-20.3	0.6	-15.3	-17.3	0.8	-12.1	11.9	1.3	10.9	0.5	3.1	260	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.7	23.5	31.4	22.9	30.1	22.2	25.1	30.1	24.3	29.1	23.6	28.1	4.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.2	27.5	22.9	18.1	26.6	22.3	17.5	26.0	77.7	29.9	74.4	28.9	71.6	28.2	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.8	DB	-20.8	34.5	4.3	1.8	-23.8	35.9	-26.3	36.9	-28.7	37.9	-31.8	39.3	(4)
(5)			WB	-21.0	26.5	4.2	0.9	-24.0	27.1	-26.5	27.6	-28.8	28.1	-31.8	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-1.9	0.6	5.0	11.5	17.4	21.8	23.6	22.8	19.1	12.7	6.2	1.2		
	DBStd	10.24	7.11	6.41	6.19	5.26	4.65	3.48	3.02	2.77	4.03	4.84	5.25	5.88		
	HDD10.0	1314	374	269	182	44	3	0	0	0	26	138	278			
	HDD18.3	2959	628	497	415	212	76	11	1	39	184	364	531			
	CDD10.0	1941	4	6	26	88	231	353	423	397	273	111	23	6		
	CDD18.3	544	0	0	1	6	46	113	165	141	62	10	0	0		
	CDH23.3	4788	0	1	11	64	412	967	1497	1186	557	92	3	0		
	CDH26.7	1571	0	0	1	6	100	327	557	383	179	17	0	0		
(14) Wind	WSAvg	3.4	4.3	4.1	4.2	4.2	3.3	2.9	2.7	2.5	2.6	3.2	3.6	3.7		
(15) Precipitation	PrecAvg	1056	78	54	79	107	117	108	116	82	84	78	69	84		
	PrecMax	1544	236	116	174	245	205	199	204	153	251	149	151	217		
	PrecMin	766	28	19	24	30	40	44	47	18	21	20	23	26		
	PrecStd	176	43	28	31	46	49	37	37	33	53	37	32	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	21.0	25.0	27.5	31.1	33.7	35.5	33.2	33.4	29.0	22.5	18.0		
		MCWB	14.5	15.1	16.3	18.2	21.9	23.2	25.0	23.6	22.3	20.0	16.5	15.0		
	2%	DB	15.0	16.3	21.8	25.3	29.1	31.7	32.8	32.0	30.9	25.9	19.5	15.2		
		MCWB	13.1	12.3	14.8	16.7	20.4	22.6	24.3	23.3	21.5	18.2	14.7	12.5		
	5%	DB	12.1	13.1	18.6	22.9	27.7	30.2	31.4	30.3	28.8	23.6	17.3	12.6		
		MCWB	10.7	10.1	13.1	15.5	19.9	21.9	23.4	22.5	20.8	17.1	13.4	10.7		
	10%	DB	8.4	10.1	15.4	20.9	25.9	28.7	30.0	28.9	26.9	21.3	14.9	10.1		
		MCWB	6.2	7.2	11.3	14.6	19.1	21.1	22.7	21.8	19.4	16.3	11.5	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.1	17.8	20.1	23.3	24.9	26.7	25.7	24.5	22.2	17.9	15.8		
		MCDB	16.5	19.7	22.7	25.9	28.7	30.2	32.3	30.2	30.3	27.0	20.5	17.7		
	2%	WB	13.3	13.5	15.7	18.4	22.1	24.0	25.6	24.8	23.3	20.3	15.9	13.3		
		MCDB	15.0	15.5	19.9	22.9	27.0	29.4	30.8	29.3	28.4	23.0	18.4	14.7		
	5%	WB	10.5	10.3	14.1	17.0	21.1	23.2	24.7	24.0	22.2	18.7	14.1	11.0		
		MCDB	11.9	12.9	17.3	20.8	25.6	28.1	29.5	28.2	26.3	22.3	16.3	12.6		
	10%	WB	6.4	7.2	11.8	15.5	20.1	22.4	23.9	23.1	21.2	16.9	12.0	8.0		
		MCDB	7.5	9.7	15.1	19.4	24.3	26.8	28.2	27.0	24.5	20.5	14.4	9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.6	12.0	11.4	10.9	10.8	11.1	12.0	11.6	9.9	8.1		
		MCDBR	10.4	13.6	14.7	14.5	13.4	12.7	12.0	12.3	13.9	14.5	13.2	11.0		
	5% WB	MCWBR	8.9	10.3	9.0	8.3	6.3	5.3	4.7	5.0	5.8	7.6	9.2	9.1		
		MCWBR	9.9	12.5	12.4	12.0	11.0	10.8	10.7	10.2	10.9	11.8	10.6	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.327	0.348	0.393	0.439	0.448	0.450	0.445	0.395	0.357	0.325	0.304		
		taud	2.440	2.342	2.365	2.254	2.166	2.193	2.193	2.189	2.309	2.410	2.457	2.455		
	Ebn at Noon		865	887	906	881	844	835	828	822	844	843	830	834		
		Edh at Noon	80	102	111	133	148	145	143	140	117	95	79	74		
(44) All-Sky Solar Radiation	RadAvg		1.70	2.56	3.56	4.80	5.49	6.07	5.94	5.46	4.46	3.13	2.10	1.44		
	RadStd		0.15	0.30	0.31	0.40	0.46	0.40	0.35	0.27	0.41	0.29	0.26	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	N/A		
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+65		

Nomenclature: See separate page